Random Walks in One Dimension

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1 Introduction

Suppose a squirrel is standing at the origin of an infinitely long 1 dimensional park with no obstructions. While waiting for the park to be developed, the squirrel decides to play a game. Every minute, the squirrel randomly decides to move one hop in either direction. What is the probability that it ends up in different locations in the park? What happens if we introduce obstacles? What if the squirrel doesn't have an equal probability of moving in each direction?

These questions can be answered by looking at random walks, which were originally developed to analyze the random motion of particles.

2 A 1 dimensional random walk with no stopping points

First, consider a random walk of length N that starts at position 0 on the number line. Since the squirrel randomly chooses a direction to move in, we can use the analogy of flipping a coin to represent the choice - heads means moving a unit in the positive direction, while tails means moving back a unit.

Note that after the sequence of N coin tosses, our squirrel will end up at the point $h-t$ on the number line, where h is the number of heads and $t = N - h$ is the number of tails in the sequence of N tosses. For example, if all N coin tosses end up as H, the squirrel will end up at position N, and conversely if all coin tosses end up as T, the squirrel will end up at position $-N$. These being the two extreme points, after N tosses the squirrel's final position will line between $-N$ and N.

2.1 Finding the probability of ending up in different positions

We are interested in characterizing $P_N(x)$: the probability that the squirrel ends up at position x after hopping N times. Since she starts at 0,

$$P_0(0) = 1$$

Suppose the squirrel only takes a single hop, the direction of which is decided by tossing a coin. If it is a head, she will end up at 1, and if it is a tail, she will end up at -1. So,

$$\begin{aligned} P_1(1) = P(H) &= \frac{1}{2} \\ P_1(0) &= 0 \\ P_1(-1) = P(T) &= \frac{1}{2} \end{aligned}$$

Similarly, after 2 hops, the probabilities are as follows:

$$\begin{aligned} P_2(2) = P(HH) &= \frac{1}{4} \\ P_2(1) &= 0 \\ P_2(0) = P(HT \cup TH) &= \frac{2}{4} \\ P_2(-1) &= 0 \\ P_2(-2) = P(TT) &= \frac{1}{4} \end{aligned}$$

And, after 3 hops the probabilities are:

$$\begin{aligned} P_3(3) = P(HHH) &= \frac{1}{8} \\ P_3(2) &= 0 \\ P_3(1) = P(HHT \cup HTH \cup THH) &= \frac{3}{8} \\ P_3(0) &= 0 \\ P_3(-1) = P(HTT \cup THT \cup TTH) &= \frac{3}{8} \\ P_3(-2) &= 0 \\ P_3(-3) = P(TTT) &= \frac{1}{8} \end{aligned}$$

Now, we notice a pattern: only a certain set of points are reachable after every N hops. For example, after 2 hops the squirrel can only end up on points $\{-2, 0, 2\}$ with positive probability while after 3 hops the squirrel can only end up on points $\{-3, -1, 1, 3\}$ with positive probability.

For any point x, we can reach x after N hops if

$$x = h - t = h - (N - h) = 2h - N, h \in \{0, \dots, N\}.$$

In other words, to reach point x we need $h = \frac{x+N}{2}$ heads, which is only possible when x has the same parity as N, as h can only take integer values. Therefore, the squirrel can only end up at points that have the same parity as the number of steps taken.

The probability of ending at point x after N hops, where $x \in \{-N, -N+2, \dots, N-2, N\}$ can thus be characterized by the probability of ending up with $h = \frac{x+N}{2}$ heads in N coin tosses. This is given by

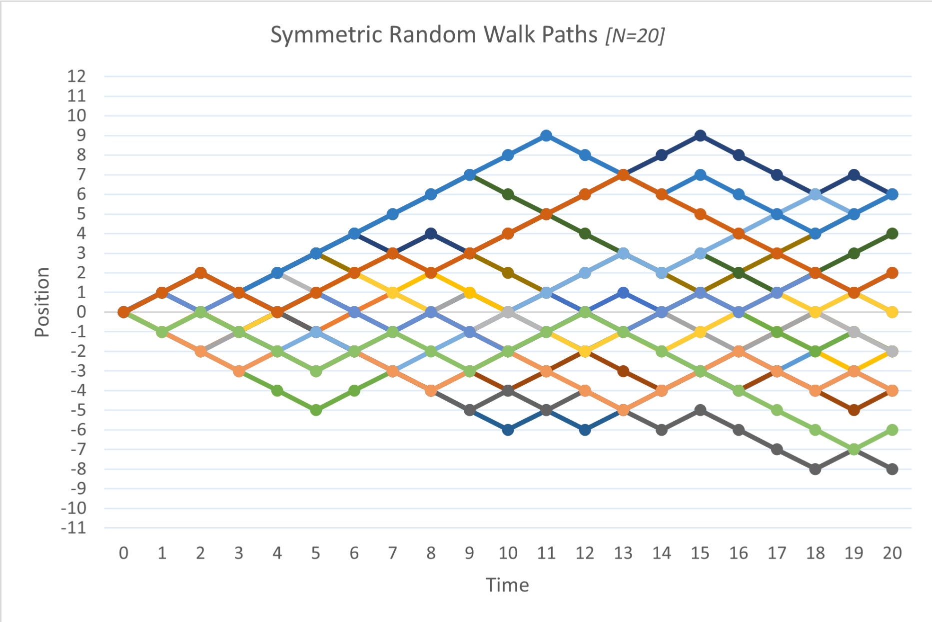
$$P_N(x) = \binom{N}{\frac{x+N}{2}} \cdot \frac{1}{2^N}.$$

The figure below illustrates a set of 20 random walk paths with 20 steps. For each step, a change in location of -1 or 1 is randomly chosen with $P(1) = \frac{1}{2}$.

2.2 Pascal's triangle

An interesting feature emerges if we observe the numerators of the probability distributions $P_N(x)$ for $N = 0, 1, 2, \ldots$. Note that all the denominators are of the form 2^N. When $N = 0$, we

Figure 1: A set of possible paths using a fair coin and 20 tosses



get single non-zero element in the numerator of $P_0(x)$, with value 1. When $N = 1$, we have two non-zero elements in the numerator for $P_1(x)$: $[1, 1]$. For $N = 2$, we have three non-zero elements in the numerator for $P_2(x)$: $[1, 2, 1]$. Below we write out the numerators in rows, with each row representing a value of N:

$$\begin{array}{lc} N = 0 & [1] \\ N = 1 & [1,1] \\ N = 2 & [1,2,1] \\ N = 3 & [1,3,3,1] \\ N = 4 & [1,4,6,4,1] \\ N = 5 & [1,5,10,10,5,1] \end{array}$$

Notice that the numerators are just the numbers in Pascal's Triangle!

To understand why the numerators of the probabilities form Pascal's triangle, we can use the example of finding the possible paths that the squirrel can follow to reach node 0 at $N = 4$, where we have $P_4(0) = \frac{6}{16}$. There are 2 ways Squirrel can reach 0 at the end of 4 hops - tossing a head from -1 at the end of 3 hops, or tossing a tail from 1 at the end of 3 hops. Now,

$$P_3(-1) = \binom{3}{\frac{-1+3}{2}} \cdot \frac{1}{2^3} = \binom{3}{1} \cdot \frac{1}{8} = \frac{3}{8}$$

and,

$$P_3(1) = \binom{3}{\frac{1+3}{2}} \cdot \frac{1}{2^3} = \binom{3}{2} \cdot \frac{1}{8} = \frac{3}{8}$$

We can calculate $P_4(0)$ as the sum of conditional probabilities of reaching 0 at step 4 from positions

-1 and 1 at the end of step 3

$$\begin{aligned} P_4(0) &= P(H) \cdot P_3(-1) + P(T) \cdot P_3(1) \\ &= \frac{1}{2} \cdot \frac{3}{8} + \frac{1}{2} \cdot \frac{3}{8} \\ &= \frac{1}{2} \cdot \frac{3+3}{8} \\ &= \frac{6}{16} \end{aligned}$$

This illustrates how we obtain the Pascal's triangle property for probabilities in a random walk. The probability of reaching any node in the given step is equal to $\frac{1}{2}$ times the sum of probabilities of reaching the 2 adjacent nodes (at ± 1 distance from target node in the prior step). So, in the numerator we get the the sum of the numerators of the two adjacent nodes in the previous row, whereas the denominator increases by a factor of 2.

3 Gates in the Park

Park developers have decided that maintaining an infinitely long park was too expensive, so they've reduced the park to have finite length and added gates to the park! They put one gate at point k, and one gate at $-k$. The squirrel returns to the origin and begins the same process. What's the probability the squirrel reaches the end of the park at one point?

3.1 The probability of leaving the park

We break down the problem of reaching k into smaller steps. First, we consider the probability that the squirrel reaches the point $x = 1$ at some point in time. Let's denote the probability as S_1. Now, starting at 0 the squirrel can either go to $+1$ with probability $\frac{1}{2}$ or go to -1 with probability $\frac{1}{2}$. From -1 the probability of reaching $x = 1$ at some point in the future can be broken down into 2 steps:

- Starting at -1 reach 0 at some point in the future. By symmetry of the random walk, we can see that this probability would be the same as the probability of starting at 0 and reaching 1 at some point - i.e., S_1
- Starting from 0 reach 1 at some point in the future, which by definition is S_1

Therefore, we can write S_1 in the following recursion

$$\begin{aligned} S_1 &= \frac{1}{2} + \frac{1}{2} \cdot S_1 \cdot S_1 \\ S_1^2 - 2S_1 + 1 &= 0 \end{aligned}$$

Solving we obtain that $S_1 = 1$.

The probability of reaching k starting from 0 can similarly be broken into k steps - reaching 1 starting from 0, followed by reaching 2 starting from 1 and so on. By symmetry, each of the steps would have a probability of S_1. So the probability S_k that the squirrel reaches point k at some point during her random walk is give by

$$S_k = S_1^k = 1^k = 1$$

Hence, the squirrel would reach k at some point during her walk with probability 1 (and by symmetry, we can show that the same result holds for $-k$), given that the squirrel is not allowed to leave the park.

3.2 The probability of reaching an endpoint within a certain amount of time

We denote the first passage time - the first time the squirrel reaches the point $x = k$ - with τ_k.

To compute the probability distribution of the time needed by the squirrel to reach an end, we make use of the reflection principle: Suppose a path reaches the point $x = m$ at some time $t < N$, where N is the number of steps the squirrel plans on taking. Then, starting from this time t, there exists a one-to-one correspondence between the paths the end up at $x > m$ and $x < m$ that can be obtained by reflecting the remaining steps of the path on the line $x = m$. The path up to time t remains unchanged. The reflection operation on the path starting from t is equivalent to replacing every subsequent $+1$ move with a -1 move and vice versa. Having reached point m at time t, for any path ending at $x = m + i$ the reflected path will end at $m - i$. Thus, for any path ending at m, its reflection will also end at m, for any path ending at $x > m$, the reflection will end at $x < m$, and for paths ending at $x < m$, the reflection will end at $x > m$.

Therefore, the probability that point m is reached at or before time $t = N$ can be computed as

$$P(\tau_m \leq N) = P_N(x = m) + 2 \cdot P_N(x > m).$$

By symmetry we know $P_N(x > m) = P_N(x < -m)$, which allows us to further simplify this to

$$\begin{aligned} P(\tau_m \leq N) &= P_N(x = m) + P_N(x > m) + P_N(x < -m) \\ &= P_N(x = m) + 1 - P_N(-m \leq x \leq m). \end{aligned}$$

Since we know how to compute $P_N(x = k)$, we can compute $P_N(-m \leq x \leq m)$ as a summation of the individual probabilities of reaching each point between $-m$ and m. This is useful as we now only have to calculate a finite number of probabilities, as opposed to trying to compute $P_N(x > m)$.

4 An unfair coin: Asymmetric Random Walks

Instead of using a fair coin to decide which way to hop next, suppose the squirrel uses a biased coin with $P(H) = p \neq \frac{1}{2}$ and $P(T) = 1 - p$. Most of our results would still hold with some simple modifications. First note that the universe of paths traversed by the squirrel has not changed (for the squirrel to end up at position x after N steps where x and N have the same parity, the coin must still land heads $\frac{N+x}{2}$ times and tails $\frac{N-x}{2}$ times, leading to an excess of $\frac{N+x}{2} - \frac{N-x}{2} = x$ heads in N coin tosses). Thus, there are still $\binom{N}{\frac{N+x}{2}}$ ways to end up at x. What changes is that the probability of each path is now $p^{\frac{N+x}{2}}(1-p)^{\frac{N-x}{2}}$:

$$P_N(x) = \binom{N}{\frac{x+N}{2}} \cdot p^{\frac{N+x}{2}}(1-p)^{\frac{N-x}{2}}$$

When $p > \frac{1}{2}$, we would expect the random walk to drift towards the right as time passes. The expected movement after every move is given by

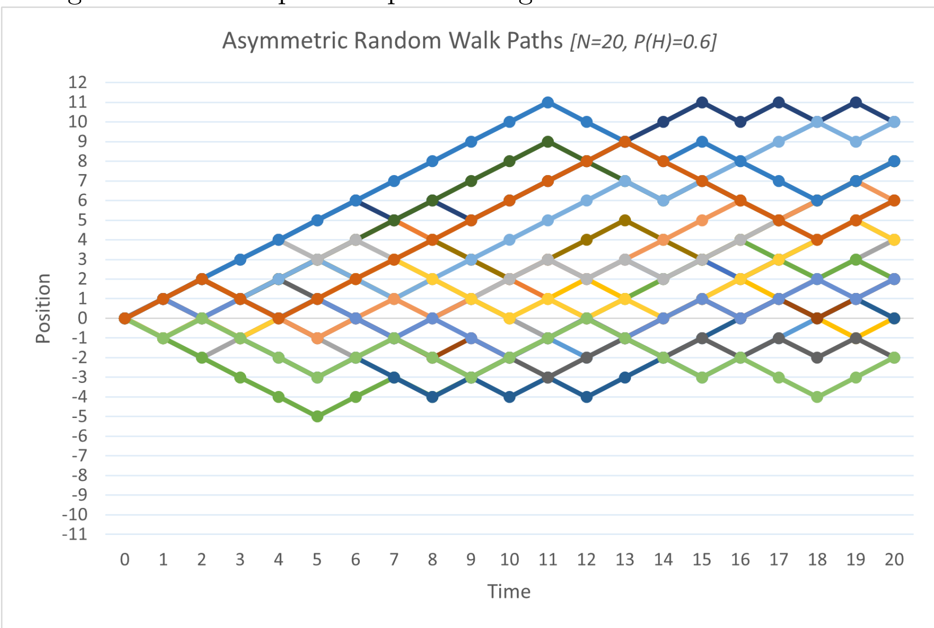
$$\begin{aligned} E[x] &= (1) \cdot p + (-1) \cdot (1 - p) \\ &= 2p - 1 \end{aligned}$$

After a N step random walk, we would expect to find the squirrel at

$$E_N[x] = N \cdot (2p - 1)$$

Whereas, with a fair coin one would expect the squirrel to end up at 0, with a biased coin the squirrel is gradually expected to move away from 0 as N increases, drifting right if $p > \frac{1}{2}$ or drifting left if $p < \frac{1}{2}$. We know this because for a single step, the expected position changes by $P(\text{heads}) \cdot 1 + P(\text{tails} \cdot (-1) = P(\text{heads}) - P(\text{tails}) = 2P(\text{heads}) - 1 = 2p - 1$, and by linearity of expectation if this is positive the path drifts right, and if it's negative the path drifts left. When $p = 1/2$ this is 0, giving us the symmetric case.

Figure 2: A set of possible paths using an unfair coin and 20 tosses



The results obtained for the probability distribution of first passage times holds as is as they contained no assumption about the fairness of the coin. Thus, even for an asymmetric random walk we can characterize the probability of first passage as

$$P(\tau_m \leq N) = P_N(x = m) + 1 - P_N(-m \leq x \leq m)$$

The probabilities $P_N(x)$ will reflect the asymmetry of the random walk.

However, intuitively it seems unlikely that under an asymmetric random walk the squirrel can reach any point k or $-k$ with equal probability. Let's reinvestigate the problem of reaching k at some point during the random walk when starting from point 0. With p denoting the probability of landing a head i.e., the next step being $+1$, we can write the recursion characterizing the probability of reaching 1 at some point during the walk as

$$S_1 = p + (1-p) * S_1^2$$
$$(1-p) * S_1^2 - S_1 + p = 0$$

We note that the sum of the coefficients of the polynomial in S_1 is 0, which implies that $S_1 = 1$ is a root of the quadratic. Then observing that the product of the roots is $\frac{p}{1-p}$, by Vieta's we obtain $S_1 = \frac{p}{1-p}$ as the other root. Thus

$$S_1 = 1, \frac{p}{1-p}$$

When $p = \frac{1}{2}$ i.e., in the case of a symmetric random walk both roots have the value of $S_1 = 1$ as we had previously obtained.

When $p > \frac{1}{2}$, we observe that one of the roots $S_1 = \frac{p}{1-p} > 1$ is degenerate as S_1 being a probability must be less than or equal to 1.

When $p < \frac{1}{2}$, we show that the root $S_1 = 1$ can be rejected. Let, $P_{N,1}$, denote the probability that the squirrel reaches 1 starting from 0 within N steps. We condition on the first step - with probability p that it reaches 1 in the next step, and with probability $1-p$ it reaches -1 and then has to reach $+1$ from -1 or equivalently 2 from 0 within $N-1$ steps. This gives

$$\begin{aligned} P_{N,1} &= p + (1-p)P_{N-1,2} \\ &\leq p + (1-p)P_{N-1,1}^2 \end{aligned}$$

The inequality obtains because the probability of moving from 0 to 1 within $N-1$ steps followed by from 1 to 2 within another $N-1$ steps ($P_{N-1,1}^2$) is greater than the probability of moving from 0 to 2 within $N-1$ steps $P_{N-1,2}$.

Note,

$$P_{1,1} = p < \frac{p}{1-p}$$

Suppose $P_{k,1} < \frac{p}{1-p}$ for some $k > 0$, Now, from the earlier inequality we obtain

$$\begin{aligned} P_{k+1,1} &\leq p + (1-p)P_{k,1}^2 \\ &\leq p + (1-p)\left(\frac{p}{1-p}\right)^2 \\ &\leq \frac{p}{1-p} \end{aligned}$$

Thus, by induction for all N, $P_{N,1} \leq \frac{p}{1-p}$. Since,

$$S_1 = \lim_{n\to\infty} P_{N,1} \leq \frac{p}{1-p}.$$

we can reject the root $S_1 = 1$ when $p < \frac{1}{2}$. In this case, $S_1 = \frac{p}{1-p}$ and $S_k = \left(\frac{p}{1-p}\right)^k$.

Combining, we obtain

$$\begin{aligned} p \geq \frac{1}{2} &\implies S_k = 1 \\ p < \frac{1}{2} &\implies S_k = \left(\frac{p}{1-p}\right)^k. \end{aligned}$$

5 Conclusion

Random walks can be used to model problems in a variety of fields! The process of random walks were initially developed to understand the motion of microscopic particles, and they are still widely used in statistical physics to model the kinetic motion and distribution of particles. In financial markets, random walks are used to model asset prices. Changes in prices of an asset is generally assumed to be independent of the past price changes. This encapsulates the idea that current markets prices incorporate all available information about the asset. Changes in asset prices are then caused by new information that are unrelated to past. Most asset prices also exhibit an upward drift over time - for example, values of stock market indices tend to go up over time. Asymmetric random walks are used for modeling such price series. The probability of up/down moves and sizes of each step are calibrated to match the properties of the price time series such as its mean drift and standard deviation. The generated random walk models can then be used to price derivative securities such as options.

References

Alm, Sven E. "Simple random walk."
math.uu.se, 9 April 2002, https://www2.math.uu.se/ sveralm/kurser/stokprocmn1/slumpvandring _eng.pdf. Accessed 29 November 2024. "Einstein's random walk."

PhysicsWorld, 15 January 2005, https://physicsworld.com/a/einsteins-random-walk/. Accessed 29 November 2024.

Fowler, Michael. "The One-Dimensional Random Walk."
phys.virginia.edu, https://galileo.phys.virginia.edu/classes/152.mf1i.spring02/RandomWalk.htm. Accessed 29 November 2024.

Johnston, Derek."An Introduction to Random Walks."
5 August 2011
https://math.uchicago.edu/ may/VIGRE/VIGRE2011/REUPapers/Johnston.pdf.

Siegrist, Kyle. 11.6: The Simple Random Walk."
Statistics LibreTexts, 23 April 2022, https://stats.libretexts.org/Bookshelves/Probability_Theory/ Probability_Mathematical_Statistics_and_Stochastic_Processes_(Siegrist)